

Smart Water Quality Monitoring System for Distribution Networks

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ABSTRACT

Drinking water quality monitoring is essential these days as the available water is polluted and can cause several diseases like cholera, diarrhea, dysentery, etc. This paper presents a low-cost wireless water quality monitoring system based on Arduino and Zigbee Module. Water quality parameters for monitoring is decided by the Central Pollution and Control Board, New Delhi, India. The water quality sensors are interfaced with the Arduino board through signal conditioning circuit. Acquired data is sent to the server over the wireless network through the Zigbee module. At the server, Raspberry Pi is used for data receiving and results have been shown on the graphical display.

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1. Introduction

Over the past decade, the water crisis has become a serious concern globally and is not limited to any country or a region. There is less water available for a human being for day by day utilization. Also, available water is polluted and can cause severe diseases like cholera, dysentery, diarrhea, etc. Contaminated drinking water is estimated to cause around five lac diarrheal deaths each year. According to a report from the World Health Organization (WHO), half of the world will be living in water stress areas by the end of 2025 (World Health Organization (WHO), 1996). In India, approx. 6% death of the people are due to diarrhea as shown in Fig. 1.

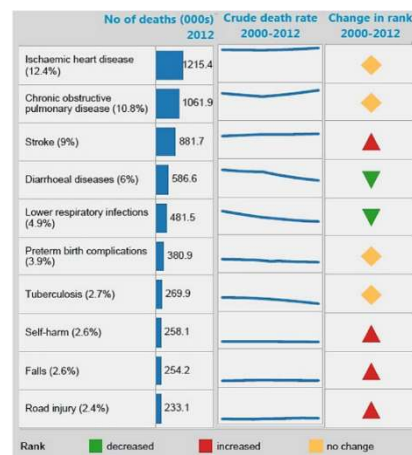


Fig. 1 - Top ten causes of death in India

These are some of the reasons for the importance of water quality monitoring. In this work, a low-cost water quality monitoring platform is designed. The water quality parameters used in this work are chosen based upon the criteria decided by Central Pollution and Control Board (CPCB), New Delhi, India (Central Pollution Control Board, 2007) and are shown in table 1. Here, we are considering 'C' category for monitoring. The water quality parameters are pH, Dissolved Oxygen (DO), Electrical Conductivity (EC), Oxidation Reduction Potential (ORP) and Temperature. A three-stage approach has been used in this paper. In the first stage, all the water quality sensors have been interfaced with the Arduino Uno board, and the data (water quality parameters) have been acquired. The acquired data is then sent to the wireless network with the help of the Zigbee module in the second stage. In the third stage, data is received, and the results have been displayed at the server end. For performance evaluation of the water quality sensors, the results have been compared with the standard conventional instrument EXO-1 purchased from YSI Incorporated, USA.

In this paper, five parameters are considered as discussed earlier and the acceptable range of the parameters are presented in Table 2. The organization of the entire paper is as follows. Materials and methods are discussed in section 2, results and discussions are described in section 3 followed by a conclusion in section 4.

Table 1 - Central Pollution and Control Board criteria for Water Quality

Type of Water	Category	Quality Parameter Criteria
Drinking Water Source without conventional treatment but after disinfection	'A'	1. Total Coliforms Organism MPN/100ml shall be 50 or less 2. pH between 6.5 and 8.5 3. Dissolved Oxygen 6mg/l or more 4. Biochemical Oxygen Demand 5 days 20°C 2mg/l or less
Outdoor bathing (Organised)	'B'	1. Total Coliforms Organism MPN/100ml shall be 500 or less pH between 6.5 and 8.5 Dissolved Oxygen 5mg/l or more 2. Biochemical Oxygen Demand 5 days 20°C 3mg/l or less
Drinking water source after conventional treatment and disinfection	'C'	1. Total Coliforms Organism MPN/100ml shall be 5000 or less 2. pH between 6 to 9 3. Dissolved Oxygen 4mg/l or more 4. Biochemical Oxygen Demand 5 days 20°C 3mg/l or less 5. TDS 2000 mg/l
Propagation of Wildlife and Fisheries	'D'	1. pH between 6.5 to 8.5 2. Dissolved Oxygen 4mg/l or more 3. Free Ammonia (as N) 1.2 mg/l or less
Irrigation, Industrial Cooling, Controlled Waste disposal	'E'	1. pH between 6.0 to 8.5 2. Electrical Conductivity at 25°C micromhos/cm Max.2250 3. Sodium absorption Ratio Max. 26 4. Boron Max. 2mg/l

Table 2 - Acceptable Range of Targeted Water Quality Parameters

S. No.	Parameters	Unit	Acceptable Range for Drinking
1	pH	pH	6.5-8.5
2	Dissolved Oxygen (DO)	mg/l	4 mg/l or more
3	Electrical Conductivity (EC)	mg/l	Less than 1000
4	Oxidation Reduction Potential (ORP)	mV	Less than 200
5	Temperature	°C	5-30 °C

2. Materials and Methods

2.1. Water Quality Parameter Selection Criteria

The selection criteria for water quality measurement is an important step as these parameters will define the water quality. As discussed earlier, these parameters are set by the CPCB, New Delhi, India and are shown in table 1. As we can observe from the table that there are various categories defined for

different application such as drinking, irrigation, and industrial purpose, etc. In this paper, type 'C' category for water quality monitoring is considered that is drinking water sources after conventional treatment and disinfection.

2.2. Hardware Development

After the water quality parameters are decided, there is a need of the hour to acquire those parameters from different water from different water samples. For this purpose, a Zigbee based low-cost platform has been developed for the demonstration. These sensors are purchased from the Atlas Scientific, USA. The block diagram of the interfacing sensors with Arduino board is shown in Fig. 2. The water quality monitoring sensors are interfaced with the Arduino board through signal conditioning circuit and serial port expander. Arduino Uno board is used as a client or sensing node, and Raspberry Pi 3 board is used as a server node.

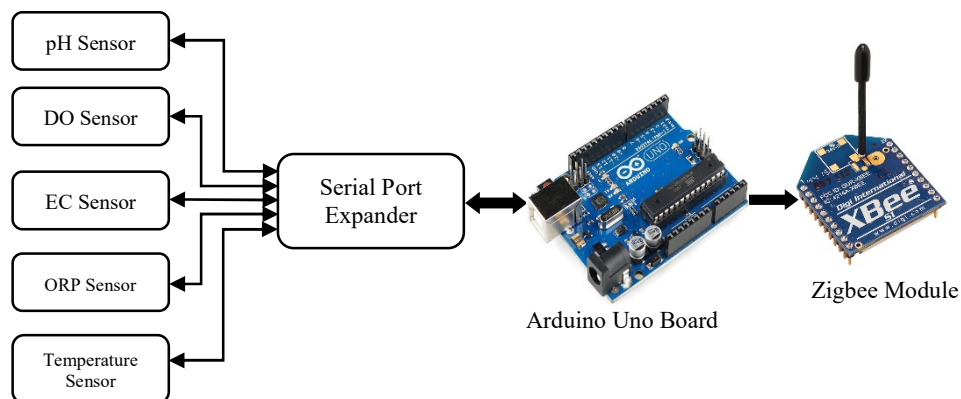


Fig. 2 – Block diagram of the sensing node

Raspberry Pi is a single board credit card size micro-computer with ARM cortex A-53 processor (Raspberry Pi Foundation, 2014). It is open hardware with many on-board running components like CPU, graphics, memory, USB controller, etc. Nowadays, Raspberry Pi board is being used in many real-time applications (for e.g. real-time paper currency recognition of new Indian notes after demonetization (Anilkumar and Srikanth, 2018), automatic traffic control system (Talukder et al., 2017), smart traffic system (Kumar et al., 2017), energy management system based on real-time electricity pricing model (Qureshi et al., 2017), self-driving system (Sumardi et al., 2018), 3D wavelet transform (Bernabé et al., 2018) (Durugkar, S. R., & Poonia, R. C. (2018), air quality monitoring system (Alkandari and Moein, 2018). Arduino is an open source electronic platform for the ease of programming. Arduino boards are ready to use boards for reading inputs, generating analog and digital outputs and publishing something online (Arduino, n.d.). A Zigbee module is interfaced with the Arduino board for wireless data transfer. The block diagram for the server node is shown in Fig. 3. A keyboard, mouse and a 7-inch screen is interfaced with the Raspberry Pi board for programming and result display.

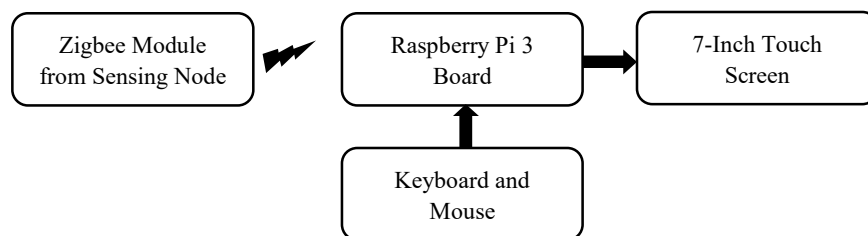


Fig. 3 – Block diagram of a server node

3. Results and Discussion

A Zigbee and Arduino based embedded platform is designed for data acquisition and data transfer. The developed hardware setup for sensing node as well as for server node is shown in Fig. 4. This developed platform is tested for various water samples, and results have been displayed on the 7-inch display.

All the water quality sensors have been calibrated before data acquisition. The water quality parameters are acquired through the Arduino board and sent over the wireless network to the server node. Different RO water samples, as well as tap water samples, have been tested, and the results are shown in table 3. Sample number 1 to 3 are RO samples, and 4 to 5 are tap water samples. The results obtained from the proposed work has been compared with the conventional instrument as mentioned in the introduction. The Percentage Relative Error (PRE) (Lee, 2016) (Durugkar, S. R., & Poonia, R. C., 2017) has been calculated for performance evaluation.

The PRE is given by the following equation. The PRE for measured samples is shown in table 4.

$$PRE = \left(\frac{Actual - Observed}{Actual} \right) * 100 \quad (1)$$

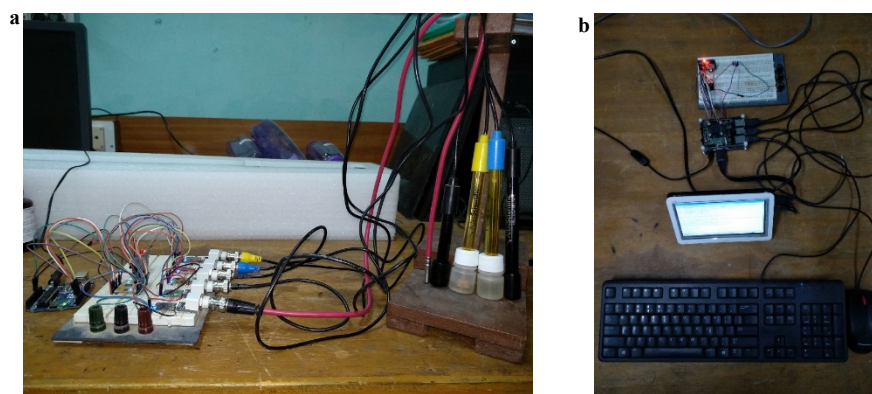


Fig. 4 – Developed Setup for (a) sensing node; (b) server node

Table 3 – Results comparison of water quality parameter from developed setup and conventional instrument

Samples	Parameters									
	pH		DO		EC		ORP		Temp	
	Developed Setup	Exo-1	Developed Setup	Exo-1	Developed Setup	Exo-1	Developed Setup	Exo-1	Developed Setup	Exo-1
Sample 1	8.2	8.35	7.8	7.6	325.4	320.5	141.2	138.8	28.6	28.45
Sample 2	8.1	8.19	7.9	7.82	128.4	125.2	134.6	132.2	28.5	28.17
Sample 3	7.7	7.77	7.8	7.73	129.2	127.7	140.8	138.9	28.8	28.63
Sample 4	8.2	8.21	7.8	7.7	1582	1576.2	170.5	168.5	29.2	28.98
Sample 5	8.1	8.04	7.5	7.4	1635	1621.8	170.9	168.7	30.1	29.84
Sample 6	8.1	8.01	7.5	7.4	1672	1662.1	171.1	168.4	30.5	30.21

Table 3 – Results comparison of water quality parameter from developed setup and conventional instrument

Samples	PRE (%)				
	pH	DO	EC	ORP	Temp
Sample 1	1.79	2.63	1.52	1.72	0.52
Sample 2	1.09	1.02	2.55	1.81	1.17
Sample 3	0.9	0.9	1.17	1.36	0.59

Sample 4	0.12	1.2	0.36	1.18	0.76
Sample 5	0.75	1.35	0.81	1.30	0.87
Sample 6	1.12	1.35	0.59	0.58	0.96

4. Conclusion

In this paper, design and development of Arduino based low-cost hardware has been presented for water quality monitoring. It has both educational and practical importance. The proposed work may be an efficient way to monitor water quality parameters in distribution networks. The proposed system comprises of water quality sensors, serial port expander, Arduino Uno board, Raspberry Pi 3 board, and a Zigbee module.

The server node has built-in Wi-Fi module, so this can be connected to the internet and IoT can be implement in the same proposed work which will be done in the future. The water quality sensors degrade over a certain period and needs timely calibration. So, in the future looks, the focus will be on drift analysis and the compensation of the sensors by means of algorithms.

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